Python Programming Assignment

1. Student Grade Management System

Topics: Dictionaries, Lists, Functions, Loops

Create a program that stores marks of multiple students in multiple subjects. The program should:

- Add a student
- Update marks
- Display report card
- Calculate average
- Display topper

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2. Library Book Management

Topics: Dictionaries, Lists, Functions

Maintain a collection of books.

Features:

- Add new book
- Borrow book
- Return book
- Search by title
- Display available books

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3. Employee Salary Analyzer

Topics: Lists, Dictionaries, Functions

Store employee details.

Calculate:

- Highest salary
- Lowest salary
- Average salary
- Employees earning above average

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4. Frequency Analysis of Words

Topics: Strings, Dictionaries

Given a paragraph:

- Remove punctuation
- Convert to lowercase
- Count frequency of every word
- Display top 5 most frequent words

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5. Matrix Operations Without NumPy

Topics: Nested Lists, Loops

Implement:

- Matrix addition
- Matrix subtraction
- Matrix multiplication
- Matrix transpose

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6. Inventory Management System

Topics: Dictionaries, Functions

Store products with:

- Name
- Quantity
- Price

Allow:

- Purchase product
- Restock
- Search
- Display total inventory value

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7. Password Strength Checker

Topics: Strings, Functions

Check whether a password satisfies:

- Minimum 8 characters
- Uppercase letter
- Lowercase letter
- Digit
- Special character

Return a strength rating.

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8. Cricket Scoreboard

Topics: Dictionaries, Lists

Store batting statistics.

Calculate:

- Highest scorer
- Strike rate
- Team total
- Average runs
- Display scorecard

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9. Banking Management System

Topics: Functions, Dictionaries

Implement:

- Create account
- Deposit
- Withdraw
- Balance inquiry
- Transfer money
- Transaction history

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10. Advanced String Compression

Topics: Strings

Example:

Input:
aaabbccccdd

Output:
a3b2c4d2

Also implement decompression.

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11. Shopping Cart Simulator

Topics: Dictionaries, Lists

Allow users to:

- Add items
- Remove items
- Update quantity
- Apply discount
- Generate final bill

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12. Election Voting System

Topics: Dictionaries, Control Flow

Create an electronic voting system.

Features:

- Register candidates
- Cast votes
- Prevent duplicate voting
- Display winner
- Show vote percentages

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13. Student Attendance Tracker

Topics: Dictionaries, Lists

Store attendance for multiple days.

Generate:

- Attendance percentage
- Students below 75%
- Highest attendance

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14. Text Statistics Analyzer

Topics: Strings

Given a paragraph, calculate:

- Characters
- Words
- Sentences
- Vowels
- Consonants
- Digits
- Special characters
- Longest word
- Shortest word

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15. Expense Tracker

Topics: Lists, Dictionaries

Store daily expenses.

Generate:

- Monthly expense
- Category-wise expense
- Highest spending category
- Average daily expense

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16. Dictionary-Based Contact Book

Topics: Dictionaries

Implement:

- Add contact
- Delete contact
- Update contact
- Search by name
- Search by phone number
- Display all contacts alphabetically

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17. Seating Arrangement System

Topics: Lists, Loops

Represent seats in a classroom.

Features:

- Reserve seat
- Cancel reservation
- Display available seats
- Count occupied seats
- Find consecutive empty seats

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18. Tournament Ranking System

Topics: Lists, Dictionaries

Store match results.

Calculate:

- Points table

- Goal difference (or score difference)
- Winner
- Runner-up
- Sort rankings

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19. Advanced Number Analyzer

Topics: Functions, Loops

For any number determine:

- Prime
- Armstrong
- Perfect
- Strong
- Palindrome
- Neon
- Automorphic
- Sum of digits
- Product of digits
- Reverse number

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20. Mini Student Database Management System

Topics: Functions, Lists, Dictionaries, Strings

Each student should contain:

- ID
- Name
- Age
- Course
- Marks (5 subjects)

Implement a menu-driven system:

1. Add Student
2. Delete Student
3. Search Student
4. Update Student
5. Display All Students
6. Display Topper
7. Sort by Name
8. Sort by Marks
9. Calculate Average Marks
10. Exit
